

ACUTE MYELOID LEUKEMIA — Clinical One-Pager

Myeloid-precursor overproliferation · untreated days–weeks, treated ~30% 5-yr · WHO/ICC/ELN 2022 · IM resident reference

1 · PATHOPHYSIOLOGY

Overproliferation of **myeloid stem cells OR myeloblasts** → disrupts RBC/WBC/plt lines. Median age 68; ~1% of cancers.

CBC pattern	Lesion
Isolated ↑WBC	Myeloblast (WBC very high or low)
↑WBC + cytopenias	Blasts crowd/disrupt other lines
Pancytopenia	Myeloid stem cell → all lines down

2 · DIAGNOSIS

Both required: (1) ≥20% blasts in marrow OR blood, (2) myeloid origin (**Auer rods**, MPO+, or immunophenotype). AML-defining lesion [t(8;21), inv(16), t(6;9), PML::RARA] = diagnostic regardless of blast count.

WHO/ICC/ELN 2022: 20% cutoff waived if AML-defining lesion present; NPM1-mut = distinct entity (≥10% blasts); MRD by NGS/flow.

Findings: WBC median 15k (>100k in 10–20%, <5k in ~25%) · anemia · thrombocytopenia (~25% <25k). Exam: gingival hyperplasia, leukemia cutis, Sweet syndrome, petechiae/pallor.

3 · FOUR CRITICAL COMPLICATIONS

Dx	Recognize / dx	Treat
Leukostasis	Neuro + resp sx; WBC >100k (or >50k+sx)	Cytoreduce (hydroxyurea/ chemo) ± leukapheresis. No leukapheresis in APL
TLS	↑K, ↑uric acid, ↑phos, ↓Ca, AKI	IVF + allopurinol; rasburicase if severe
DIC	↑PT/PTT/INR, ↑D-dimer, ↓fibrinogen, schistocytes	Transfuse (cryo/plt/FFP) + treat leukemia; APL → ATRA/arsenic
Febrile neutropenia	T >38.3 (or ≥38×1h) + ANC <500	Cefepime or pip-tazo STAT; +vanc only if line/ SSTI/PNA/unstable

Pearls: APL → NO leukapheresis (worsens coagulopathy). Avoid acetaminophen in neutropenic inpatients — masks fever.

4 · APL — THE ONE NOT TO MISS

Promyelocytes + Auer rods + DIC. Start **ATRA** on suspicion (don't wait for PML::RARA). Tx = ATRA + arsenic. Excellent prognosis — **acute phase is the deadliest window**.

5 · RISK (ELN) & PROGNOSIS

By cytogenetics + NGS:

Group	5-yr OS
Favorable	~46%
Intermediate	~18%
Adverse	~4%

NGS (FLT3, NPM1, IDH1/2, TP53, CEBPA) drives prognosis AND targeted therapy.

6 · TREATMENT FRAMEWORK

Systemic only (no surgery/RT → no neoadj/adj language).
• **Induction** (treat detectable dz): 7+3 (7d cytarabine + 3d anthracycline); Ven/HMA if old/unfit; ATRA/arsenic for APL
• **Consolidation** (treat undetectable dz): allogeneic SCT or chemo
• **Maintenance:** oral azacitidine (Onureg); **R/R** → salvage/ targeted vs hospice

2022–25 targeted/maintenance: quizartinib+7+3 (FLT3-ITD, QuANTUM-First); ivosidenib+aza (IDH1, AGILE); olutasidenib (R/R IDH1); CPX-351 (t-AML/AML-MR).